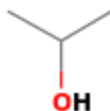


Isopropyl Alcohol

- **Formula:** C₃H₈O
- **Molecular weight:** 60.0950
- **IUPAC Standard InChI:** InChI=1S/C3H8O/c1-3(2)4/h3-4H, 1-2H3  **InChI**
v 1.06
- **IUPAC Standard InChIKey:** KFZMGEQAYNKOFK-UHFFFAOYSA-N 
- **CAS Registry Number:** 67-63-0
- **Chemical structure:**



This structure is also available as a [2d Mol file](#) or as a [computed 3d SD file](#)

[View 3d structure](#) (requires JavaScript / HTML 5)

- **Other names:** 2-Propanol; sec-Propyl Alcohol; Alcojel; Alcosolve 2; Avantin; Avantine; Combi-Schutz; Dimethylcarbinol; Hartosol; Imsol A; Isohol; Isopropanol; Lutosol; Petrohol; Propol; PRO; Takineocol; 1-Methylethyl Alcohol; iso-C₃H₇OH; 2-Hydroxypropane; Propane, 2-hydroxy-; sec-Propanol; Propan-2-ol; i-Propylalkohol; Alcolo; Alcool isopropilico; Alcool isopropylique; Alkolave; Arquad DMCB; iso-Propylalkohol; Isopropyl alcohol, rubbing; IPA; Lavacol; Visco 1152; Alcosolve; i-Propanol; 2-Propyl alcohol; Spectrar; Sterisol hand disinfectant; UN 1219; n-Propan-2-ol; 1-methylethanol; Propanol-2; Virahol; IPS 1
- **Permanent link** for this species. Use this link for bookmarking this species for future reference.
- **Information on this page:**
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- **Other data available:**

- [Condensed phase thermochemistry data](#)
- [Phase change data](#)
- [Reaction thermochemistry data](#)
- [Henry's Law data](#)
- [Gas phase ion energetics data](#)
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- [IR Spectrum](#)
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- **Data at other public NIST sites:**
 - [Gas Phase Kinetics Database](#)
 - [X-ray Photoelectron Spectroscopy Database, version 5.0](#)
- **Options:**
 - [Switch to calorie-based units](#)

Data at NIST subscription sites:

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Gas phase thermochemistry data

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Data compiled as indicated in comments:

[ALS](#) - Hussein Y. Afeefy, Joel F. Liebman, and Stephen E. Stein

[DRB](#) - Donald R. Burgess, Jr.

[GT](#) - Glushko Thermocenter, Russian Academy of Sciences, Moscow

Quantity	Value	Units	Method	Reference	Comment
$\Delta_f H^\circ_{\text{gas}}$	-272.8	kJ/mol	Eqk	Buckley and Herington, 1965	ALS
$\Delta_f H^\circ_{\text{gas}}$	-271.1	kJ/mol	N/A	Chao and Rossini, 1965	Value computed using $\Delta_f H^\circ_{\text{liquid}}$ value of -317.0 ± 0.3 kJ/mol from Chao and Rossini, 1965 and $\Delta_{\text{vap}} H^\circ$ value of 45.9 kJ/mol from Snelson and Skinner, 1961. ; DRB
$\Delta_f H^\circ_{\text{gas}}$	-272.3 ± 0.92	kJ/mol	Ccb	Snelson and Skinner, 1961	ALS
$\Delta_f H^\circ_{\text{gas}}$	-272.8	kJ/mol	N/A	Parks, Mosley, et al., 1950	Value computed using $\Delta_f H^\circ_{\text{liquid}}$ value of -318.7 kJ/mol from Parks, Mosley, et al., 1950 and $\Delta_{\text{vap}} H^\circ$ value of 45.9 kJ/mol from Snelson and Skinner, 1961. ; DRB

Constant pressure heat capacity of gas

$C_{p,\text{gas}}$ (J/mol*K)	Temperature (K)	Reference	Comment
35.32	50.	Thermodynamics Research Center, 1997	p=1 bar. Discrepancies with other statistically calculated values [Green J.H.S., 1963] and [51KOB] increase at high temperatures up to 5 and 9 J/mol*K, respectively, in $C_p(T)$. There is a good agreement with results [Chao J., 1986]. Please also see Chao J., 1986, 2. ; GT
46.04	100.		
57.98	150.		
68.28	200.		
83.72	273.15		
89.32 ± 0.15	298.15		
89.74	300.		
112.15	400.		

131.96	500.
148.30	600.
161.75	700.
173.04	800.
182.67	900.
190.97	1000.
198.16	1100.
204.41	1200.
209.85	1300.
214.60	1400.
218.75	1500.
227.0	1750.
233.1	2000.
237.6	2250.
241.0	2500.
243.7	2750.
245.7	3000.

Constant pressure heat capacity of gas

C _{p,gas} (J/mol*K)	Temperature (K)	Reference	Comment
103.06	358.72	Stromsoe E., 1970	Ideal gas heat capacities are given by [Stromsoe E., 1970] as a linear function $C_p=f_1^*(a+bT)$. This expression approximates the experimental values with the average deviation of 1.59 J/mol*K. The accuracy of the experimental heat capacities [Stromsoe E., 1970] is estimated as less than 0.3%. Other experimental values of C_p [Parks G.S., 1940] (118.83 at 427.9 K, 127.61 at 457.7 K, and 135.56
105.7 ± 1.6	365.75		
105.77	371.15		

106.29	373.15
108.1 ± 1.6	378.85
109.2 ± 1.6	384.95
110.08	391.15
110.8 ± 1.6	393.65
111.65	398.15
113.0 ± 1.6	405.35
114.35	411.15
117.02	423.15
118.70	431.15
122.10	448.15
122.80	451.15
121.7 ± 1.6	453.15
124.2 ± 1.6	466.75
127.01	473.15
126.7 ± 1.6	480.55
130.3 ± 1.6	499.75
132.9 ± 1.6	513.95
137.5 ± 1.6	539.05
142.6 ± 1.6	567.05
148.1 ± 1.6	597.25

J/mol*K at 480.3 K) are believed to be less reliable. Please also see [Hales J.L., 1963](#), [Berman N.S., 1964.](#); *GT*

References

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- Symbols used in this document:

$C_{p,gas}$ Constant pressure heat capacity of gas

$\Delta_f H^\circ_{gas}$ Enthalpy of formation of gas at standard conditions

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